

SplitPriLoc: Split Learning for Privacy Preserving Wi-Fi Based Indoor Localization

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Abstract—Traditional Wi-Fi-based indoor localization systems face significant privacy risks when raw Channel State Information (CSI) or model updates are shared with external servers. While Federated Learning frameworks mitigate some of these concerns, they remain vulnerable to membership inference and gradient inversion attacks, and they are notoriously ill-suited for resource-constrained client hardware. To address these challenges, we propose SplitPriLoc, an indoor localization framework that integrates split learning into an existing encoder-decoder localization pipeline. By dividing the neural network architecture during training, SplitPriLoc computes compact embeddings on local Access Points (APs) and transfers only low-dimensional smashed data to a cloud server, ensuring raw data remains local while lifting the heavy computational burden off edge hardware. We evaluated SplitPriLoc against the open-source FedWiLoc benchmark using a subset of the DLoc dataset across complex indoor environments. Experimental results demonstrate that SplitPriLoc matches the localization accuracy of federated baselines, while providing superior defense against structural privacy attacks and accommodating hardware constraints.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Indoor positioning is a foundational capability for smart buildings, asset tracking, emergency response, and a growing class of IoT and wireless sensor network (WSN) services [1]. Since GPS does not work indoors, Wi-Fi has become a practical alternative as the infrastructure is already widely deployed.

Wi-Fi access points can measure how a signal travels through a space capturing reflections, delays, and angles, then producing what is known as Channel State Information (CSI). When visualized, CSI forms a heatmap that reveals where a user is standing with sub-meter accuracy [1]. Deep neural networks trained on these heatmaps achieve state-of-the-art localization [2].

The problem is that sending this raw data to a server also exposes sensitive details about the user, such as their movement patterns and behavior, creating a serious privacy risk [3]. Simply keeping the data on the device helps, but even sharing model updates during training can leak information [2].

Most implementations of Privacy-Preserving Wi-Fi based indoor localization utilized Federated Learning. This, however, does not consider environments with resource-constrained hardware. We propose SplitPriLoc, which integrates split

learning on existing indoor localization systems to solve this critical issue.

SplitPriLoc builds upon the open source wireless localization system FedWiLoc. It aims to replicate the results of FedWiLoc while being friendly towards resource-constrained hardware. Furthermore, the implementation of Split Learning makes the system more resilient towards Membership Inference Attacks (MIA) and Gradient Inversion Attacks (GIA), which are both a vulnerability of Federated Learning [CITE].

II. RELATED WORK

A. Wi-Fi Fingerprinting and CSI Localization

Early indoor localization systems used Received Signal Strength Indicator (RSSI) values to estimate user position based on signal attenuation between APs. Although easy to implement, their accuracy was limited by multipath and interference, with median localization errors often exceeding 2-5 meters [1]. CSI-based methods recover fine-grained phase and amplitude information across subcarriers and antennas, enabling sub-meter accuracy. Three-dimensional extensions of CSI fingerprinting such as [3] apply tensor decomposition and recurrent neural networks on CSI tensors built from all OFDM subcarriers, achieving robust 3-D mobile target localization on commodity Wi-Fi hardware across both line-of-sight and NLoS conditions.

B. Federated Learning for Indoor Localization

Federated learning (FL) has emerged as a privacy-centered paradigm by enabling decentralized training across multiple clients without sharing raw data. FeMLoc [2] combines FL with model-agnostic meta-learning to produce a global meta-model for RSSI-based localization tasks, achieving up to 80.95

C. Split Learning and Privacy-Preserving Localization

FedWiLoc [1] directly addresses the vulnerability to architectural privacy by combining split learning with federated averaging across APs. Each AP runs a ResNet34 encoder locally and transmits only a compact embedding to the user device, which decodes its own location corresponding to how GPS receivers compute position locally. By federating encoder training across APs using gradient averaging, FedWiLoc prevents AP-specific encoder fingerprints that could enable user tracking [1]. Our work implements and validates this core split learning design on the publicly available DLoc dataset,

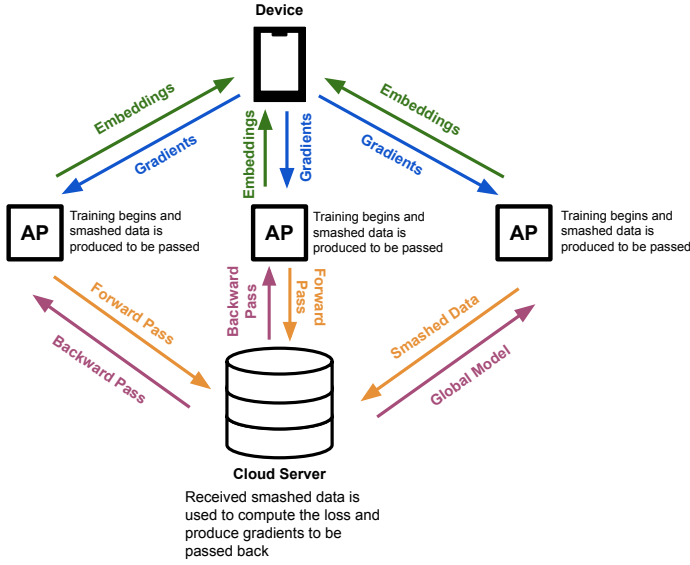


Fig. 1. Represents Split Learning in the System Model

benchmarking it against a centralized MLP and an RSSI k-NN baseline.

III. PAPER CONTRIBUTIONS

SplitPriLoc adds 3 main contributions to existing privacy-preserved localizations systems:

- Implements Split Learning instead of Federated Learning.
- Performs better against MIAs and GIAs.
- Being resource-constrained hardware friendly.

IV. SYSTEM MODEL

SplitPriLoc builds directly on the architectural foundations established by FedWiLoc [CITE], retaining the core encoder-decoder split, the AoA-ToF image representation, and the geometric loss function. The key contribution of SplitPriLoc is replacing federated learning with split learning as the training paradigm, which changes how and where the model is trained while leaving the deployment architecture unchanged.

A. Model Architecture

The model architecture of SplitPriLoc is identical to that of FedWiLoc. Each Access Point (AP) runs a local ResNet34-based encoder that processes AoA-ToF images derived from Channel State Information (CSI) measurements and produces a compact 64-dimensional embedding vector. These embeddings are transmitted to the user device, which aggregates them and passes them through a lightweight MLP decoder to predict one Angle-of-Arrival (AoA) value per AP. The predicted angles are then combined with the known physical positions of the APs in a least-squares triangulation step to produce a final (x, y) location estimate. Raw CSI data never leaves the AP at any point, preserving the core privacy guarantee of FedWiLoc. The

difference between SplitWiLoc and FedWiLoc lies entirely in how this model is trained.

B. Split Learning

In split learning, the model is split across two physical machines during training; In the case of SplitPriLoc, the AP and the cloud server. Rather than training a self-contained model entirely on one machine, the AP trains only a portion of the model, produces smashed data to be forwarded to the cloud server, and then the cloud server trains the remaining portion. Training is therefore a collaborative process between the AP and the cloud server on every batch. [Figure 1](#)

The training protocol proceeds as follows:

- Each AP encoder processes its local feature map and produces smashed data in the cut layer.
- The smashed data are forwarded to the server. item
- The server computes AoA predictions, confidence scores, and the final 2-D location via weighted ray intersection.
- The total loss is computed on the server and backpropagated
- Gradients at the cut layer flow back to each AP encoder

V. PROPOSED SCHEME

SplitPriLoc implements the equations and algorithms used by FedWiLoc, this was chosen to keep the focus of the results primarily on the Split Learning change.

A. Algorithm

SplitPriLoc utilizes the FedAvg algorithm used by FedWiLoc. Algorithm 1's description was taken straight from the paper which describes the algorithm in more detail.

Algorithm 1 FedAvg. The K clients are indexed by k ; B is the local minibatch size, E is the number of local epochs, and η is the learning rate.

```

initialize  $w_0$ 
for each round  $t = 1, 2, \dots$  do
     $m \leftarrow \max(C \cdot K, 1)$ 
     $S_t \leftarrow$  (random set of  $m$  clients)
    for each client  $k \in S_t$  in parallel do
         $w_{t+1}^k \leftarrow (k, w_t)$ 
    end
     $m_t \leftarrow \sum_{k \in S_t} n_k$ 
     $w_{t+1} \leftarrow \sum_{k \in S_t} \frac{n_k}{m_t} w_{t+1}^k$  // Erratum4
end
 $(k, w)$ : // Run on client  $k$   $\mathcal{B} \leftarrow$  (split  $\mathcal{P}_k$  into batches of size  $B$ )
for each local epoch  $i$  from 1 to  $E$  do
    for batch  $b \in \mathcal{B}$  do
         $w \leftarrow w - \eta \nabla \ell(w; b)$ 
    end
end
return  $w$  to server

```

TABLE I
OVERVIEW OF EVALUATION DATASETS

Abbr.	Scenario	Location	Space Size	No. of APs	Description
E1	July 28th	Jacobs Hall	18m × 8m	4	Complex high-multipath and NLOS environment (1500 sq. ft.).
E2	Aug. 16th-1	Jacobs Hall	18m × 8m	4	Complex high-multipath and NLOS environment (1500 sq. ft.), with extra furniture placed randomly.

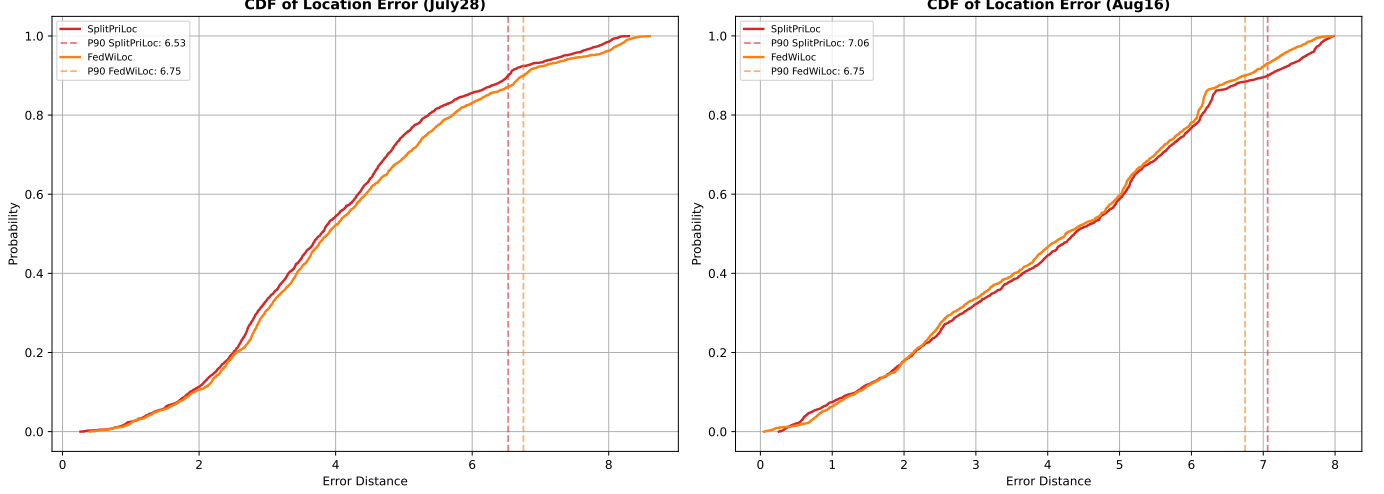


Fig. 2. Cumulative Distribution Function (CDF) of Location Error for July 28 and August 16. SplitPriLoc outperforms FedWiLoc for the July28 dataset while the opposite is true for the Aug16 dataset.

B. Equations

A Geometric Loss Function is utilized for direct angle predications in our system. It was proposed by FedWiLoc and it decomposes angles into trigonometric components:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \lambda_1 \cdot \text{Huber}(\cos(\theta_{\text{gt}}), \cos(\theta_{\text{pred}}), \delta_1) \\ & + \lambda_2 \cdot \text{Huber}(\sin(\theta_{\text{gt}}), \sin(\theta_{\text{pred}}), \delta_2) \\ & + \lambda_3 \cdot \text{Huber}(\mathbf{X}_{\text{gt}}, \mathbf{X}_{\text{pred}}, \delta_3) \end{aligned} \quad (1)$$

VI. IMPLEMENTATION

A. System Implementation

SplitPriLoc's system implementation was derived from FedWiLoc's open-sourced code base titled "PriWiLoc". SplitPriLoc switches out the Federated Learning model with a Split Learning model while keeping the architecture of the rest of the pipeline the same.

B. Hyperparameter Tuning and Training

To attempt to replicate the results from FedWiLoc, the following parameters have been applied:

- Adam Optimizer has been used
- Learning rate was set to 5×10^{-5}
- Batch size was set to 32
- The model has been trained to a maximum of 50 epochs

C. Dataset

SplitPriLoc uses the same dataset that FedWiLoc uses. It provides an excellent starting framework for later evaluation. FedWiLoc uses both the DLoc and RLoc datasets. In the context of this project, only a subset of the DLoc dataset has been used: Jacobs Hall July 28th and Aug 16th scenarios. Table I provides an overview of the chosen datasets.

VII. RESULTS AND EVALUATION

Using the results from the FedWiLoc paper directly did not make sense in the context of this project. Comparably, FedWiLoc has a much bigger dataset and environments that comparing results with the current implementation of SplitPriLoc does not show the full picture. To validate the results of SplitPriLoc, we reran the FedWiLoc pipeline with the shortened dataset we had to work with. This allowed fair and accurate validation of our system.

A. Evaluation Parameters

The evaluation parameters that have been chosen are based on the testing metrics generated by both FedWiLoc and SplitPriLoc, which include:

- 90th Percentile Location Error Figure 2
- AoA Prediction Figure 3
- Location Error over Samples Figure 4
- Predicted Location Tracking Figure 5

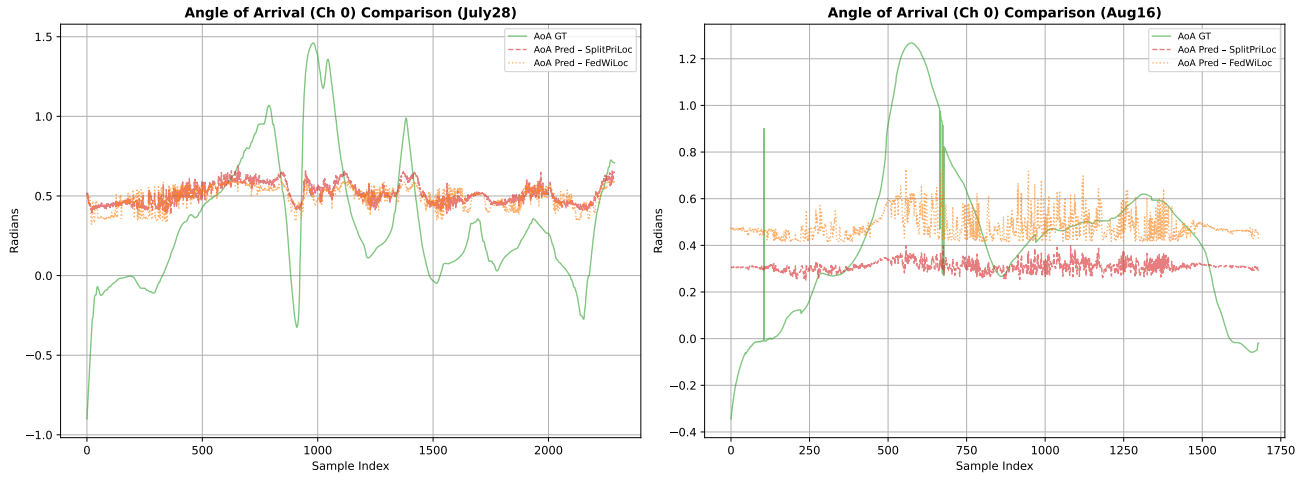


Fig. 3. Angle of Arrival Comparison between Ground Truth, SplitPriLoc, and FedWiLoc. While the predictions are not very accurate, both SplitPriLoc and FedWiLoc yield similar results.

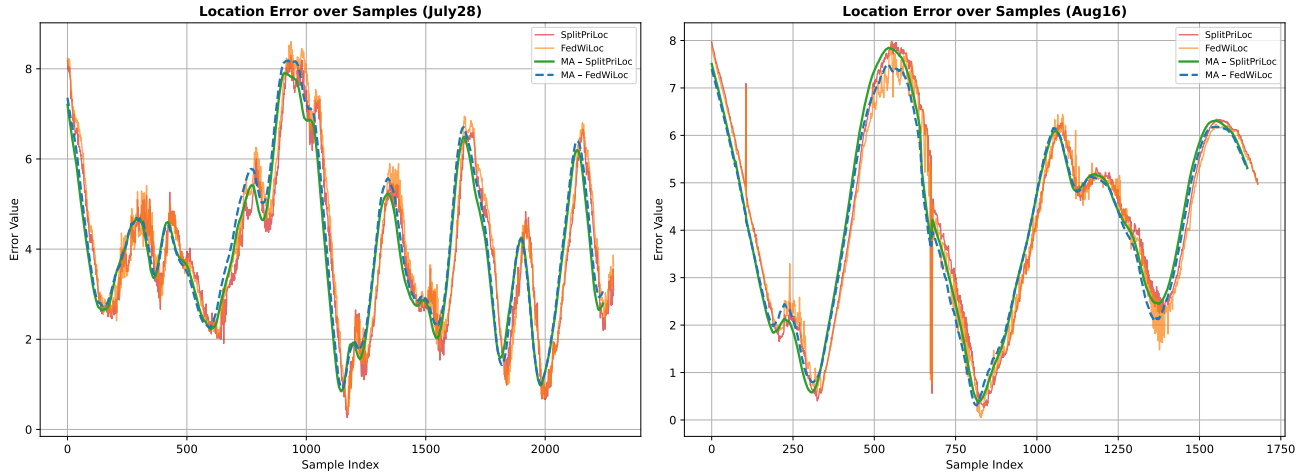


Fig. 4. Location Error and Moving Averages (MA) tracked across sample indices. Both SplitPriLoc and FedWiLoc yield great performance in the evaluation of this metric.

B. Results

Results showed that SplitPriLoc and FedWiLoc yield very similar outcomes. In E1, SplitPriLoc was able to outperform FedWiLoc by a tiny margin, while for E2, SplitPriLoc fell short and FedWiLoc outperformed SplitPriLoc by a tiny margin. With these current results, we can infer that SplitPriLoc can perform just as well as FedWiLoc with the added benefits mentioned previously, like the ability to run SplitPriLoc on resource-constrained hardware.

VIII. FUTURE WORK

A. Challenges

The project in this context faced a few challenges, mainly:

- The inability to use the full dataset
- The inability to exactly replicate results from the original FedWiLoc paper due to the inability of using the full dataset and environment presented

B. Further Improvements

There are a lot of improvements that can be done if we want to publish this paper:

- Attempt to improve the implementation of FedWiLoc to make it more tailored for Split Learning.
- Attempt to utilize a different algorithm that could be tailored better towards Split Learning.
- Deploy the system in an environment to compare the actual results from the main FedWiLoc paper.
- Compare generalization of the system with FedWiLoc.
- Measure SplitPriLoc's effectiveness on resource-constrained hardware.

IX. CONCLUSION

We proposed SplitPriLoc, a privacy-preserving Wi-Fi indoor localization system that replaces federated learning with a split learning paradigm to secure Channel State Information (CSI) data. By decoupling the model at a designated cut layer, the

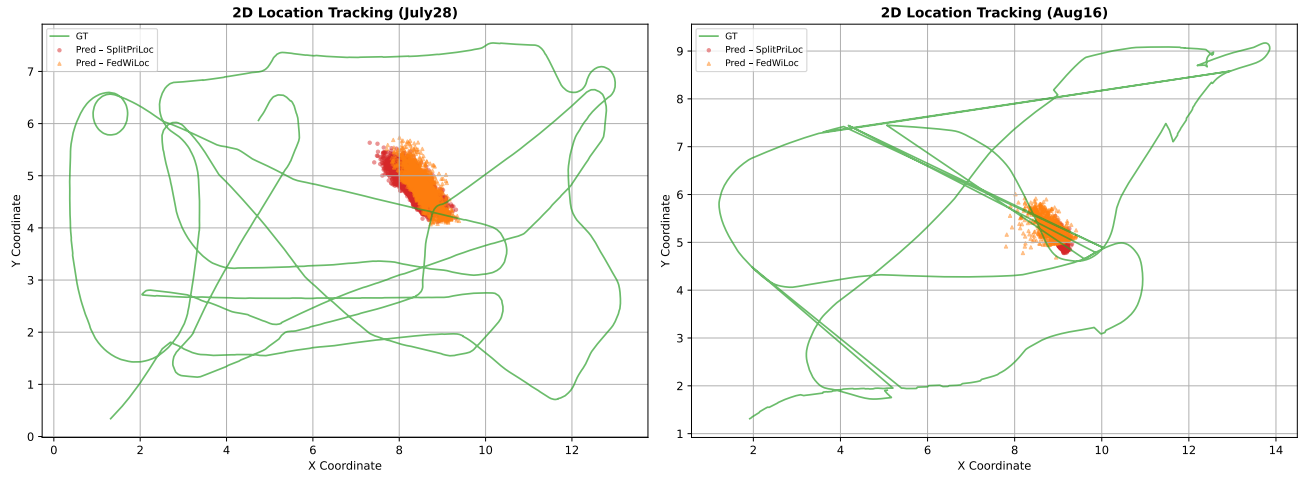


Fig. 5. 2D Location Tracking trajectories showcasing Ground Truth alongside predicted model coordinates. While results seem way off, both SplitPriLoc and FedWiLoc seem to yield similar predictions.

architecture processes features locally on Access Points before transmitting compact embeddings to a centralized cloud server for triangulation.

The evaluation outcomes indicate that SplitPriLoc yields performance highly comparable to the FedWiLoc baseline across the tested environments. Specifically, SplitPriLoc achieved slightly lower localization error distances in the July 28th complex scenario, while falling marginally behind in the August 16th furniture-modified setup.

The main lesson learned is that splitting model training significantly reduces the local computational footprint without compromising the system’s baseline positioning capabilities. Furthermore, restricting server interaction to post-cut layer smashed data inherently minimizes the risk of structural gradient and membership leakage.

REFERENCES